LIB

Reg. No. | | | | | | | | | | |

[17CE41]

FOURTH SEMESTER B.TECH EXAMINATION, JUNE – 2019

BUILDING CONSTRUCTION AND CONCRETE TECHNOLOGY

Time: 3 Hours **Maximum Marks:70**

Instructions:

i. Answer any **FIVE** full Questions choosing atleast any **TWO** full questions from each part.
ii. Assume any missing data suitably if necessary
iii. Draw neat diagrams wherever required

PART - A

1.	a.	What do you mean by load bearing and framed structures?	04
	b.	Enumerate and explain various building materials.	10
2.	a.	Define plastering. Explain the method of three coat lime plastering.	08
	b.	State and Explain types of pointing.	06
3.	a.	Write note on,	
		i. Louvered windows	03
		ii. Sliding Doors	03
		iii. Swing doors	03
	b.	List and explain the defects in timber	05
4.	a.	Briefly explain the factors affecting selection of flooring materials.	06
	b.	Write an explanatory note on pitched roof.	08

PART - B

5.	a.	Explain the manufacturing process of cement by wet process with help of flow chart.	07
	b.	Summaries the procedure of conducting consistency test on cement.	07
6.	a.	Discuss the factors affecting workability of concrete.	07
	b.	With the help of neat graph explain the relation between strength and water/cement ratio of concrete.	07
7.	a.	State different types of mineral admixtures. Explain GGBS as admixture in detail.	08
	b.	Write brief note on ,	
		i. Fiber reinforced concrete.	03
		ii. Polymer concrete.	03

JAIN (Deemed-to-be University) Learning Resource Centre Jain Global Campus Ramanagar-112

8. Calculate the mix proportions for M20 grade concrete for following stipulations of proportioning. **14**

DESIGN STIPULATIONS:

Grade designation: M20

Type of cement: OPC 43 grade conforming IS 8112

Maximum nominal size of aggregate: 20mm

Minimum cement content: 320 kg/m^3

Maximum Cement content: 450 kg/m^3

Maximum W/C ratio: 0.55

Workability: 75mm(Slump)

Exposure condition: Mild

Degree of supervision: Good

Type of aggregate: crushed angular aggregate

Chemical admixture: Not recommended

TEST DATA OF MATERILAS:

Cement used OPC 43 grade conforming IS 8112

Specific gravity of Cement 3.15

Coarse aggregate 2.68

Fine aggregate 2.65

Water absorption of Coarse aggregate 0.6%

Fine aggregate 1%

Surface moisture of Coarse aggregate and Fine aggregate: NIL

Sieve analysis of Coarse aggregate: Conform to table 3 of IS383

Fine aggregate: Zone II